Hacettepe University | Computer Engineering
CMP 720 EMBEDDED SYSTEMS DESIGN

Spring 2026

Midterm Project Peer Evaluation Form

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Instructions

For each project, briefly summarize the work and provide your evaluation **individually (not as a group)**. Your response should reflect your understanding of the problem, methodology, and evaluation plan, and include critical insights, strengths, limitations, and possible improvements. Skip your own group's evaluation and simply state that it is your group.

Students are required to evaluate only the projects presented within their own class group (Hacettepe or ASELSAN).

Guiding points (address in one coherent paragraph):

- What problem is being solved, why is it important, and what are the most important embedded system design considerations for this problem (e.g., timing, reliability, energy, safety, security, scalability, etc.)?
- What is the main idea of the proposed methodology?
- What do you think are the strengths and limitations of the approach?
- Is there anything you would improve or do differently?
- What is the most critical or challenging component?
- Is the evaluation plan sufficient? What, if anything, is missing?

Group 1

Umut Kalman, Eren Karadağ

Project Title: *Context-Aware Deadline Scheduling Under Memory Contention for Embedded Real-Time Systems*

This project addresses the problem of sporadic deadline misses caused by "hidden" memory contention, where tasks sharing a memory bus interfere with each other. It is important because standard schedulers (like EDF) assume perfect hardware isolation, which fails in real deployment and causes timing jitter. Key design considerations include runtime determinism, minimal overhead, and safety.
The main idea is a lightweight "Slack-Aware" EDF wrapper that reorders tasks based on their memory intensity without requiring heavy hardware changes.
The most challenging component is the penalty model, which must accurately predict memory interference to make reordering effective. The evaluation plan is sufficient, using physical hardware (STM32L4) and comparing results against a theoretical baseline.

Group 2

Yusuf Çağrı Öğüt

Project Title: *Hybrid Cellular-LoRa Network Based Agriculture Monitoring and Control Application*

Not evaluated as it is my project.

Group 3

Yusuf Tahir Orhan

Project Title: *Implementation of DeFT: A Deadlock-Free and Fault-Tolerant Routing Algorithm*

The DeFT algorithm aims to solve the critical challenges of deadlocks and hardware failures in modular 2.5D chiplet architectures. This is important because the vertical connections between chips are highly prone to physical defects. The design prioritizes reliability, resource efficiency, and predictable timing. Its main idea is a fault-tolerant routing strategy using three strict Virtual Network rules that prevent data from getting stuck in loops. To improve the approach, the presenter plans to replace these fixed tables with Reinforcement Learning to dynamically optimize data paths, as also planned for master's thesis. The most challenging component is the 5-stage synchronous pipeline, which must ensure that all data movements stay perfectly timed within a single clock cycle.

Group 4

Mehmet Yusuf Sezgi, Rana Elif Albayrak

Project Title: *Real-Time CAN Decoder*

This project aims to solve the problem of decoding messy vehicle data (CAN bus traffic) into useful information in real-time, which is vital for car diagnostics and performance monitoring. The design prioritizes timing and reliability to ensure no data is lost even when the vehicle generates heavy traffic. One of the biggest challanges in project is there is no access to real DBC files. The study differentiates from others as it exhibits real-time embedded implementation and lightweight coding. Its main idea is a two-stage buffering system that quickly grabs incoming data and saves it in a temporary storage area to prevent bottlenecks. While the testing plan is solid, it could be improved by checking how the system handles real world physical conditions during extensive working.

Group 5

Metin Irkıçatal, Anıl Arda

Project Title: *Edge-AI Based Smart Access Control System*

Group 6

Oğulcan Uğuroğlu, Barış Büyükyılmaz

Project Title: *Contention-Aware Duplication Scheduling for NoC Systems*

Group 7
Ahmet Yiğitalp Demirci

Project Title: *Real-Time Energy-Constrained UAV Optimization*

Group 8
Sami Enes Yılmaz, Kadir Emre Avcı

Project Title: *SmartAquaria System*

Group 9
Mehmethan Ayrım

Project Title: *Energy-Aware Mixed-Criticality Scheduling*

Group 10

İrem Gül Subaşı

Project Title: *Fault Injection and Resilience Profiling of Lightweight Transformers*

Group 11

Süleyman Ege Karakaya

Project Title: *Adaptive Data-Movement Strategy for Crypto Accelerators*

Group 12

Gökçer Erol

Project Title: *Acoustic Localization System for Search and Rescue*

This project is very meaningful due to it's purpose. They propose ryhtmic acuoustic signals for rescue missions. Basically by using diferent sound receiver sources at different locations, sounds coming from survivor will be received. There are many challanges to project. One is even microseconds delay is crucial as sound travels really fast. Another one is as the system will work on a disaster area, power probably will not be present. Another strong side of the project is it uses ML in embedded systems. Evaluation plan is also solid.

Group 13

Emre Doğan, Yavuz Selim Kızıltaş

Project Title: *Optimization of Edge AI Pipeline on TDA4VM*

Group 14

Mehmet Yusuf Bilge, Sezil Ağırbaş

Project Title: *Closed-Loop Vestibular Stimulation System*

This project is very meaningful as it proposes a solution to preventable falls which may result in even deaths. While existing GVS systems are often lab-bound and expensive, this project proposes a wearable, real-time, closed-loop device that uses postural feedback to assist with balance. Using sensor fuzion and kalman filter in embedded system is very challenging. Battery and latency requirements are also very challenging, which is the more crucial requirement.

Group 15

Ayberk Aygün, Berkay Taşkın

Project Title: *Quadrotor Simulation on NVIDIA Jetson*

The project is exciting as it deals with a common real-world problem. As there are many factors which effect the system, it is a good idea to increase system robustness by applying potential disturbances via monte carlo simulations and getting the best possible result from these scenarios. Nevertheless, using six seperate controllers and trying to tune them by this method is not very efficient. It would be a better approach use optimal control technique such as model predictive control.

Group 16

Yasin Arslan, Süleyman Sertaç Üst

Project Title: *Real-Time Audio Pipeline Analysis on Heterogeneous SoCs*

Group 17

Oğuz Kağan Yağlıoğlu, Emre Esef Kılıçdoğan

Project Title: *Predictive Maintenance of BLDC Motors via Multi-Sensor Fusion*

The project is meaningful as it points to a real world problem., which has different application fields. The work is strong as it deals with sensor fusion. Sensor layout plan is very strong. Using ML in this battery-powered and resource limited system is also a very big challange. The evaluation plan is very strong.

Group 18

Hüseyin Eren Doğan

Project Title: *EdgeRAG: Offline RAG on Embedded Systems*

Group 19

Ali Bölücü

Project Title: *Spectral-Seeded NSGA-II for Many-Core Systems*

The project focuses on the multi-core application mapping problem in a Network-on-Chip (NoC) environment. Mapping applications to specific cores is an NP-hard problem. Poor mapping results in long routing distances, high latency and etc. The project is challenging as it inherits both embedded system development and optimization. The methodology is really strong and evaluation plan is efficient.

Group 20

İlter Doğaç Dönmez

Project Title: *Autonomous Obstacle Avoidance Vehicle using LiDAR*

The project idea is strong and valuable as testing and finding out problems before production is very important due to economic concerns, and when the field is autonomy also safety is very crucial. The design prioritizes reliability and safety in low-luminosity environments by relying exclusively on a single LIDAR sensor rather than CV techniques. The project is challenging as it inherits different actuators and controller modules.